A MULTI-VIEW RETRIEVAL SYSTEM

MULTI-FACETED MUSIC RETRIEVAL.

Combining audio, text, and joint embedding into a single recommendation system that outperforms any single view.

Team

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Dataset

Free Music Archive · 2,000 tracks

/ 01 — THE PROBLEM

A single embedding can't answer every kind of music question.

QUERY · MOOD

"Chill lo-fi vibes"

requires acoustic features — mood & instrument feel

QUERY · SEMANTIC

"Folk artists about travel"

requires text & lyrics — semantic meaning

QUERY · STRUCTURAL

"Songs like this one"

requires graph relationships — co-genre, co-artist

No single embedding captures all three.

→ build independent views, then fuse

/ 02 — ARCHITECTURE

From raw audio & metadata to fused recommendations.

This project builds a **multi-view retrieval system** over the Free Music Archive (FMA) dataset, an open-licensed benchmark of 8,000 tracks across 8 top-level genres. Each view produces an independent embedding space, and a fusion layer combines them to outperform any single view.

/ INPUTS

- MP3 audio · 30s clips
- tracks.csv · genres.csv
- Genius API · lyrics

/ EMBEDDING MODELS

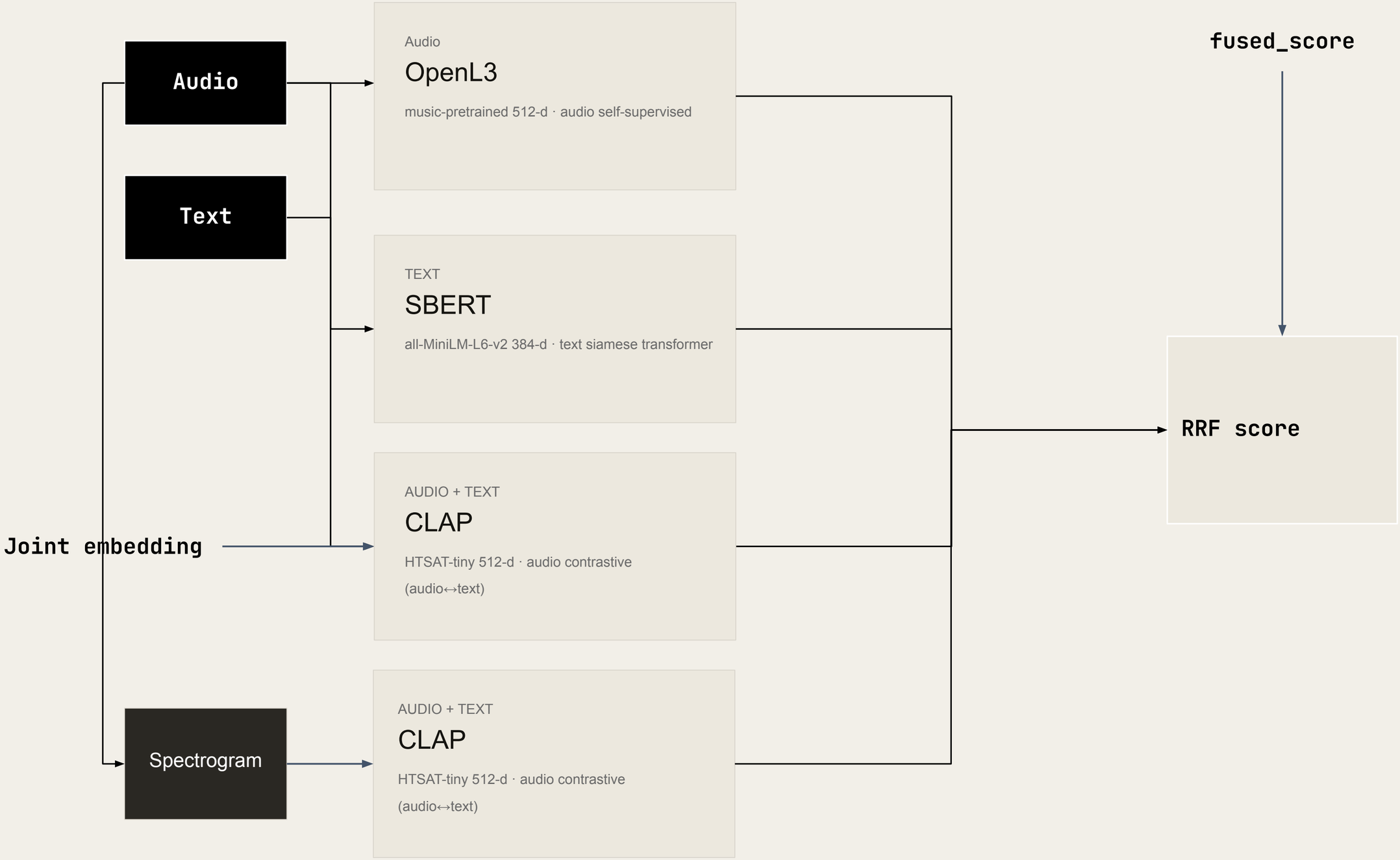
- CLAP · HTSAT-tiny — 512-d
- SBERT · all-MiniLM-L6-v2 — 384-d
- OpenL3 · self-supervised — 512-d

256-d

/ FUSION & OUTPUT

- Score fusion · weighted

L2-normalised vectors · cosine via dot product


Audio
Text
Joint embedding
Spectrogram
Audio
OpenL3
music-pretrained 512-d · audio self-supervised
TEXT
SBERT
all-MiniLM-L6-v2 384-d · text siamese transformer
AUDIO + TEXT
CLAP
HTSAT-tiny 512-d · audio contrastive
(audio↔text)
AUDIO + TEXT
CLAP
HTSAT-tiny 512-d · audio contrastive
(audio↔text)
fused_score
RRF score

MULTI-VIEW MUSIC RETRIEVAL

Search title or artist

BROWSE TRACKS

All | **Hip-Hop (250)** | Folk (250) | Pop (250) | Rock (250) | Experimental (250) | International (250) | Electronic (250) | Instrumental (250)

This World AWOL HIP-HOP	The Tension Fanatic HIP-HOP	Planet Say (featuring Faust) Food For Animals HIP-HOP	Asylum (Permanent Underclass) Dälek HIP-HOP	Culture for Dollars Dälek HIP-HOP	3 Rocks Blessed Dälek HIP-HOP
Music EPMD HIP-HOP	Bad Weather Height With Friends HIP-HOP	Escape Tune Height With Friends HIP-HOP	The Robot's Heel BrokeMC HIP-HOP	The Discovery BrokeMC HIP-HOP	In the Dark BrokeMC HIP-HOP
Fucking Exotic Bitches The Professional Savage HIP-HOP	aint no thing BOPD HIP-HOP	2.18.05 dai BOPD HIP-HOP	1.26 beat1 BOPD HIP-HOP	Now Get Busy Beastie Boys HIP-HOP	No Meaning No (feat. Fine Arts Militia) Chuck D HIP-HOP
Hippie Robot Instrumental {6th Sense, Wildabeast, Jelani, Scheme} 6th Sense HIP-HOP	The Proposal Instrumental {Jelani, 6th Sense, Wildabeast} 6th Sense HIP-HOP	...Both Nice Instrumental {Both Nice} 6th Sense HIP-HOP	Ate Ten Instrumental {Both Nice} 6th Sense HIP-HOP	D'Evils 2008 Talib Kweli HIP-HOP	Belief feat. John Mayer 6th Sense & Mick Boogie HIP-HOP
comedi humaine WatFatMan HIP-HOP	court metrage WatFatMan HIP-HOP	l'air du temps live WatFatMan HIP-HOP	Is It Too Late For Us All? MUTE HIP-HOP	Time (Reprise) (Closing Credits) Virtu-oso HIP-HOP	No Mas Tha Silent Partner HIP-HOP
Moy Moy Moy Ghostown HIP-HOP	New Pirates (featuring DJ HDL) Sun Zoo HIP-HOP	110% (radio edit) Laws HIP-HOP	Rain Bilal Bashir & Divine Styler HIP-HOP	Requiem for Stateway Thaione Davis HIP-HOP	Angular Momentum Pre HIP-HOP
God Damn Sleaze HIP-HOP	Dig Deep Mix Lasswell HIP-HOP	Sorry Comfort Fit HIP-HOP	Je Suis Le Peuple Sans Visage Featuring Arnaud Michniak DJ /rupture HIP-HOP	government funded weed Black Ant HIP-HOP	Self Explainitory Black Ant HIP-HOP
5 Piece Black Ant HIP-HOP	True Form Comfort Fit HIP-HOP	Whenever You Wanna Call Me Comfort Fit HIP-HOP	Remember Something I Forgot Comfort Fit HIP-HOP	Miles of Smiles Comfort Fit HIP-HOP	Dro Please Cvees HIP-HOP
Villaintime Madness Deal The Villain HIP-HOP	Accordion Cullah HIP-HOP				

LOAD MORE

NOW PLAYING

Culture for Dollars

Dälek · HIP-HOP

0:00 / 0:00

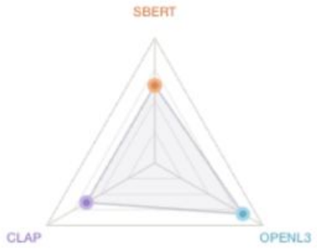
FUSED RECOMMENDATIONS

Reciprocal Rank Fusion

1	Asylum (Permanent Underclass)	Dälek · HIP-HOP	0.0487
2	Mediaocrity	The Impossebulls · HIP-HOP	0.0396
3	The Discovery	BrokeMC · HIP-HOP	0.0386
4	Escape Tune	Height With Friends · HIP-HOP	0.0320
5	Wheels Of Fortune (411)	The Dread · HIP-HOP	0.0313
6	Ghost Baby	The Crypts! · ROCK	0.0310
7	B07 Ya Dumb Bitch	Mom Jeans · POP	0.0280
8	Vandalise	Heavy Links · HIP-HOP	0.0271
9	World of Pain	Opio · HIP-HOP	0.0258
10	STOCKPILE! (Funky Dubb Mixx)	LOWdown · HIP-HOP	0.0251

Asylum (Permanent Underclass)

Dälek



0.622	0.817	0.633
SBERT	OPENL3	CLAP

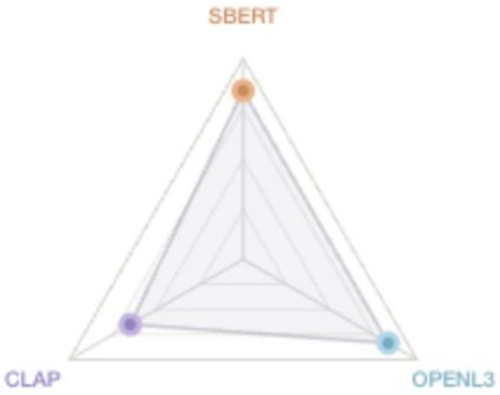
● SBERT ● OPENL3 ● CLAP

SBERT | OPENL3 | CLAP

1	3 Rocks Blessed Dälek	0.661
2	Asylum (Permanent Underclass) Dälek	0.622
3	Woman KINGS OF THE CITY	0.394
4	Zombies Pot-C	0.377
5	Please Tell Me KINGS OF THE CITY	0.376
6	Wheels Of Fortune (411) The Dread	0.376
7	Stay Fly	0.371

Here With You

Ed Askew



0.841 SBERT

0.836 OPENL3

0.650 CLAP

● SBERT ● OPENL3 ● CLAP

	SBERT	OPENL3	CLAP
1	Here With You Ed Askew		0.841
2	Piano 2 Ed Askew		0.672
3	My Dream Ed Askew		0.662
4	Sing Sing Eddi Front		0.499

/ 03 — DATASET

Free Music Archive — open-licensed benchmark.

2,000
TRACKS · FMA SMALL

8
TOP-LEVEL GENRES

294
ECHO NEST OVERLAP

/ GENRES

Electronic · Experimental · Folk · Hip-Hop · Instrumental · International · Pop · Rock

/ SOURCE

github.com/mdeff/fma · Defferrard et al., 2017

/ ADDITIONAL DATA SOURCES

FMA	tracks.csv, genres.csv — artist, title, tags, genre labels
GENIUS	Song lyrics — first 1,000 characters per track
ECHO	8 continuous audio features — danceability, energy, valence…
EVAL	294-track Echo Nest overlap → independent ground truth

/ SECTION 01

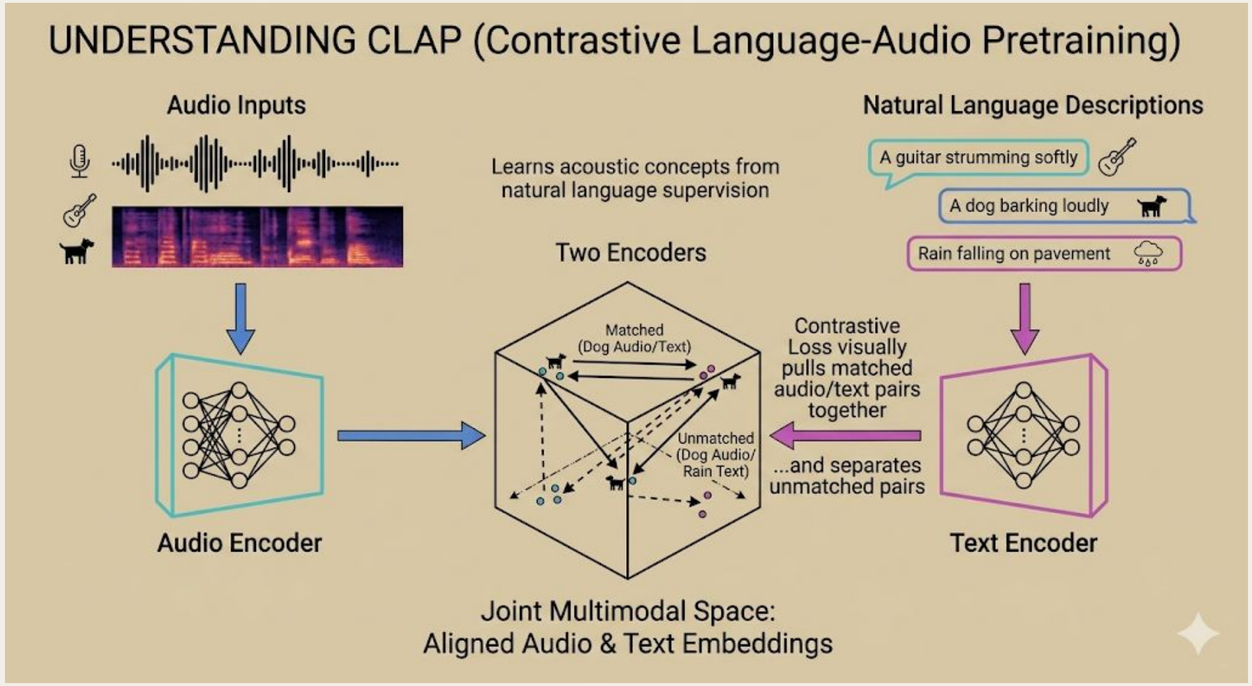
CLAP.

Vibe & text-to-music search — a 512-d shared space for audio and text via contrastive pretraining.

/ 01 — CLAP RESULTS

CLAP - Contrastive Language Audio-Pretraining

- Learns acoustic concepts from natural language supervision
- Two encoders
- Contrastive loss
- Audio and text descriptions -> joint multimodal space



CLAP retrieves on-genre top results across all natural-language queries — even with no fine-tuning.

/ 01 — CLAP RESULTS

Natural-language queries against 7,997 audio embeddings.

QUERY	TOP RESULT	GENRE	COS
"sad piano ballad"	DUITA — XPURM	Instrumental	0.65
"aggressive heavy metal"	Dead Elements — Angstbreaker	Rock	0.54
"upbeat happy pop song"	One Way Love — Ready for Men	Pop	0.52
"acoustic guitar folk"	Wainiha Valley — Mia Doi Todd	Folk	0.49

/ GENRE STRUCTURE IN CLAP SPACE

SIM Hip-Hop ≈ Pop · 0.81
Folk ≈ International · 0.80

DIFF Rock ↔ Electronic · 0.64

PCA 50 components capture ~85% of variance

CLAP retrieves on-genre top results across all natural-language queries — even with no fine-tuning.

/ SECTION 02

SBERT.

Lyrics & semantic search — a siamese transformer mapping metadata + lyrics to 384-d embeddings.

/ 02 — SBERT ANALYSIS

A more uniform 384-d space than CLAP's.

/ MODEL

all-MiniLM-L6-v2 — a 6-layer transformer with 384-d output, trained on 1B+ sentence pairs (NLI + paraphrase).

/ PCA VARIANCE

4.1%
TOP COMPONENT

181
COMPONENTS FOR 90%

vs. 50 components for CLAP's 512-d space — SBERT uses its dimensions more uniformly, which is desirable for retrieval.

/ REPRESENTATION TESTS

SEM **Semantic robustness** — only 10% overlap between "lonely" vs "isolated" queries

LEX **Lexical bias** — "Blue music" retrieves titles with "Blue", not the Blues genre

TRC **Truncation** — minimal impact; FMA strings < 50 words

/ METHODOLOGICAL CORRECTION

A leakage path from genre labels into SBERT inputs — caught and fixed.

PROBLEM

Genre leaked through input strings.

"{artist} - {title}. Tags: {genre}, {tags}"

48.8% of tags contained genre-like labels (folk, pop, melodic…). When evaluation uses genre as ground truth, the model matches on the label itself — inflating retrieval scores.

FIX

Three-step cleanup.

01 Removed genre_top from input entirely

02 Filtered all 8 genre words from tags

03 Added Genius API lyrics (1,000 chars) as semantic signal

"{title} by {artist}. Tags: {filtered}. Lyrics: {lyrics}"

→ Genre retained for evaluation only. Never seen by the embedding model.

/ SECTION 03

OPENL3.

Acoustic similarity — self-supervised audio embeddings capturing timbre, rhythm, and texture.

/ 03 — OPENL3

A purely acoustic space — no semantic supervision.

/ MODEL

OpenL3 (content_type='music' , 512-d). Trained via self-supervised audio-visual correspondence on AudioSet — captures low-level acoustic properties without any label supervision.

/ KEY DIFFERENCE FROM CLAP

CLAP	maps audio into a space shared with text — semantic
L3	maps audio into a purely acoustic space — sounds-alike
→	Two songs that sound similar are close — even across genres

USE CASE

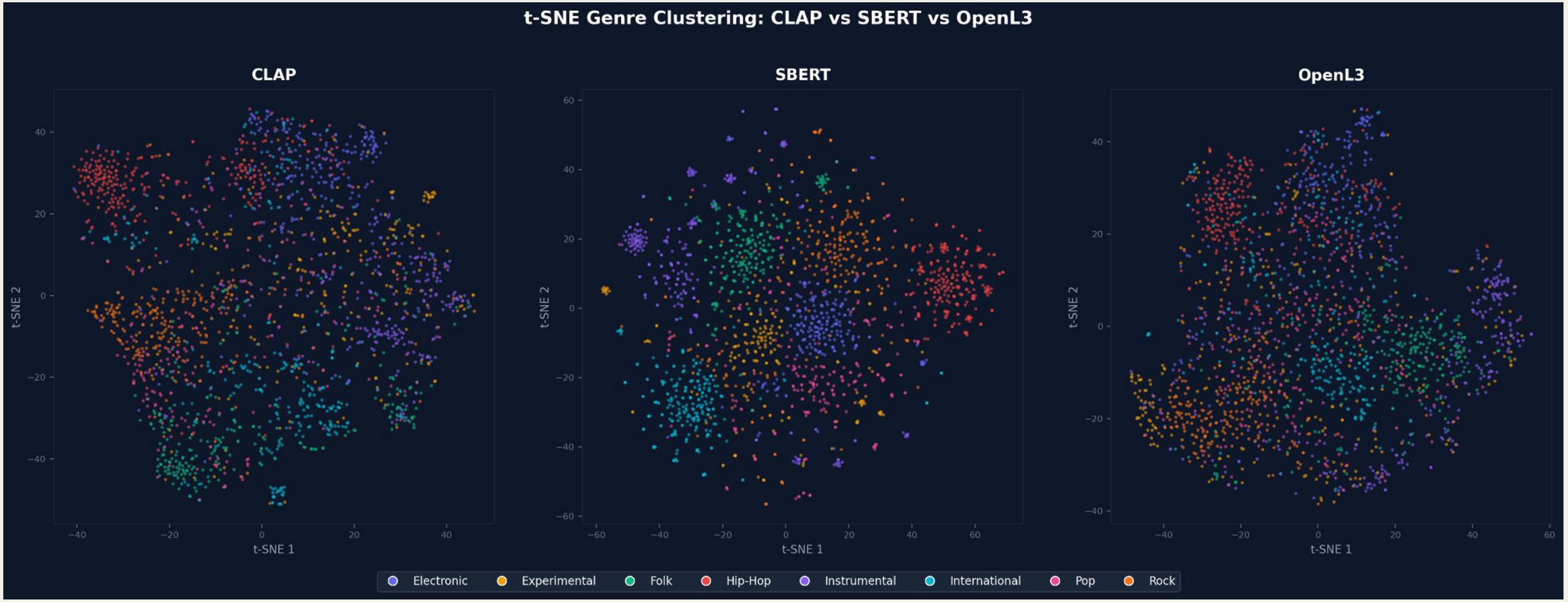
"Find songs that sound like this one."

Index all 8,000 tracks via FAISS for instant nearest-neighbour search.

2,000	512	<1ms
TRACKS INDEXED	DIMENSIONS	PER QUERY

/ 06 — WHY FUSION/Cross-View Overlap

SBERT and OpenL3 retrieve almost entirely different tracks for the same query.



/ 06 — WHY FUSION/Cross-View Overlap

SBERT and OpenL3 retrieve almost entirely different tracks for the same query.

5.6%

OVERLAP @ TOP-20

Only 1.1 of 20 neighbours shared.

-0.77

SPEARMAN RANK CORRELATION

SBERT-similar tracks are OpenL3-dissimilar.

/ INTERPRETATION

Strong complementarity — each modality contributes information the other lacks.

This is the core motivation for multi-view fusion: combining independent signals produces better retrieval than any single view alone.

/ 06 — WHY FUSION/Cross-View Overlap

Fusion Method: RRF

/ INTERPRETATION

RRF calculates a **final score** for each document using the formula:

`score(d) = Σ 1 / (k + rank(result(q), d))`

Where:

- `d` is a document in the result set
- `result(q)` is the ranked list of documents for query `q`
- `rank(result(q), d)` is the position of document `d` in that list (starting from
- k is a constant (=60) that smooths the influence of rank differences

Documents that appear high in multiple lists accumulate higher scores, while documents appearing in only one list receive lower scores. This approach **requires no training data** and is insensitive to the quality variations of individual input lists.

/ 08 — EVALUATION

Genre Retrieval Accuracy: Supervised ground truth — 2,000 tracks, 8 balanced genres, top 1 recommendation

METHOD	AVG DIST.	VS. RANDOM
CLAP · Audio	53.7%	+41.2%
OpenL3	55.3%	+42.8%
SBERT	57.6%	+45.1%
CLAP · Text	**67.7%**	**+55.2%**
Fused · SBERT+OpenL3+CLAP	**76.9%**	**+64.4%**

/ KEY TAKEAWAYS

01 CLAP text embedding is strongest single view

02 Text semantics > acoustic features for genre

03 Fused model performs significantly better than four models alone

/ 08 — EVALUATION

Echo Nest Feature: Independent ground truth — 294 tracks the models never saw, top 5 recommendations

METHOD	AVG DIST.	VS. RANDOM	P-VALUE
Random baseline	3.80	—	—
SBERT	3.33	−12.4%	3e−07
Fused · SBERT+OpenL3+CLAP	**2.87**	**−24.3%**	**1.77e-18**
OpenL3 & CLAP	**2.87**	**−24.3%**	**7e−19**

/ KEY TAKEAWAYS

01 OpenL3 & CLAP wins, because they have audio-related information.

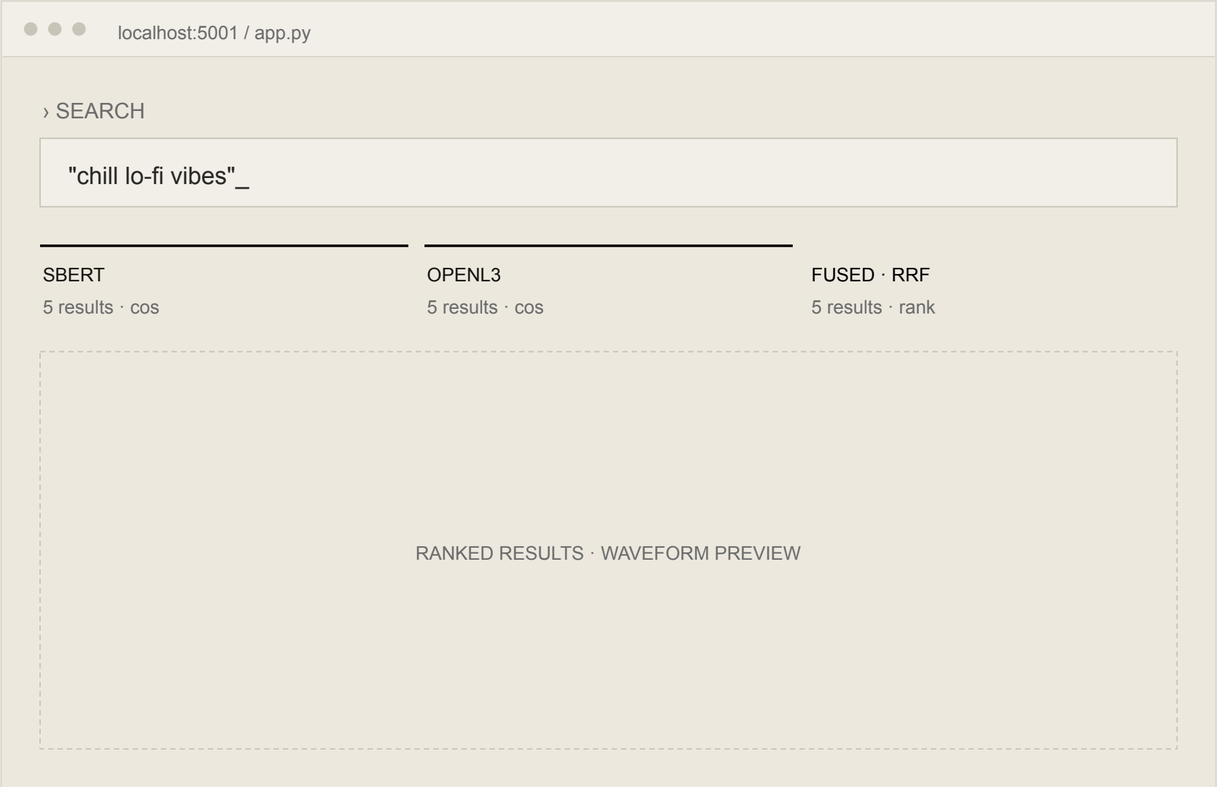
02 Fused keeps the same with OpenL3 & CLAP, but is more robust (covering more tracks)

03 Even SBERT (text only) significantly beats random ($p < 10^{-6}$)

04 All differences statistically significant (paired t-test)

/ 10 — DEMO

Multi-view recommendation app — side-by-side, per-view.



/ FEATURES

- → Search any track by title, artist, or genre
- → Per-view recommendations (SBERT · OpenL3 · CLAP)
- → Fused results via Reciprocal Rank Fusion
- → Per-view cosine scores shown alongside
- → Audio playback of original MP3 files

/ TECHNICAL

All embeddings L2-normalised; cosine via dot product. FAISS IndexFlatIP — exact search, <1ms / query @ 8k vectors.

/ 11 — FINDINGS

Four takeaways from the project.

01

Multi-view fusion outperforms any single view.

Fused: −15.1% vs random · text alone −12.4% · audio alone −24.3%.

02

Text and audio are highly complementary.

Only 5.6% top-20 overlap; Spearman ρ = −0.77.

03

Data leakage is a real risk in retrieval evaluation.

48.8% of tags contained genre labels — caught and removed before evaluation.

04

Domain ground truth requires care.

Echo Nest features provide independent, audio-derived ground truth no model saw during training.

/ APPENDIX A — GLOSSARY

Reference terms used throughout.

EMBEDDING
A fixed-size numerical vector representing a track in a continuous space where similar items are close.

COSINE SIMILARITY
Cosine of the angle between two vectors, [-1, 1]. Equals dot product for L2-normalised vectors.

CLAP
Contrastive Language-Audio Pretraining — shared audio/text space; mood, genre, instruments.

OPENL3
Self-supervised audio embeddings on AudioSet — timbre, rhythm, texture (no labels).

RRF
Reciprocal Rank Fusion — sum 1/(k+rank) across views. Robust, no weight tuning.

T-SNE
t-distributed Stochastic Neighbor Embedding — 2D visualisation of high-dimensional vectors.

FAISS
Facebook AI Similarity Search — efficient nearest-neighbour search. We use IndexFlatIP (exact inner product).

CONTRASTIVE LEARNING
Learns to map similar pairs close, dissimilar pairs far. Typically uses InfoNCE loss.

SBERT
Sentence-BERT — siamese transformer for sentence embeddings. all-MiniLM-L6-v2.

GNN
Graph Neural Network — node embeddings from neighbour aggregation. SAGEConv, heterogeneous graph.

DATA LEAKAGE
Eval targets leaking into model inputs, inflating metrics. We removed genre from SBERT input.

INFONCE
Contrastive loss using softmax over (positive vs many negatives). Used for CLAP fine-tuning.

/ APPENDIX B — REFERENCES

Datasets, models & libraries.

/ DATASETS

FMA **Free Music Archive (Small)** 8,000 tracks · 8 genres · ~30s clips · open-licensed. Defferrard et al., 2017 · github.com/mdeff/fma

ECHO **Echo Nest / Spotify audio features** Danceability · energy · valence · tempo · acousticness · … 294-track FMA overlap → independent ground truth

GEN **Genius API** Song lyrics (first 1,000 chars per track).

AS **AudioSet** Training data for OpenL3 (audio-visual correspondence). Gemmeke et al., 2017 · research.google.com/audioset

/ MODELS & LIBRARIES

CLAP **LAION CLAP** · HTSAT-tiny backbone, AudioSet+music pretraining

laion/clap

SBERT **Sentence-Transformers** · all-MiniLM-L6-v2 · 384-d

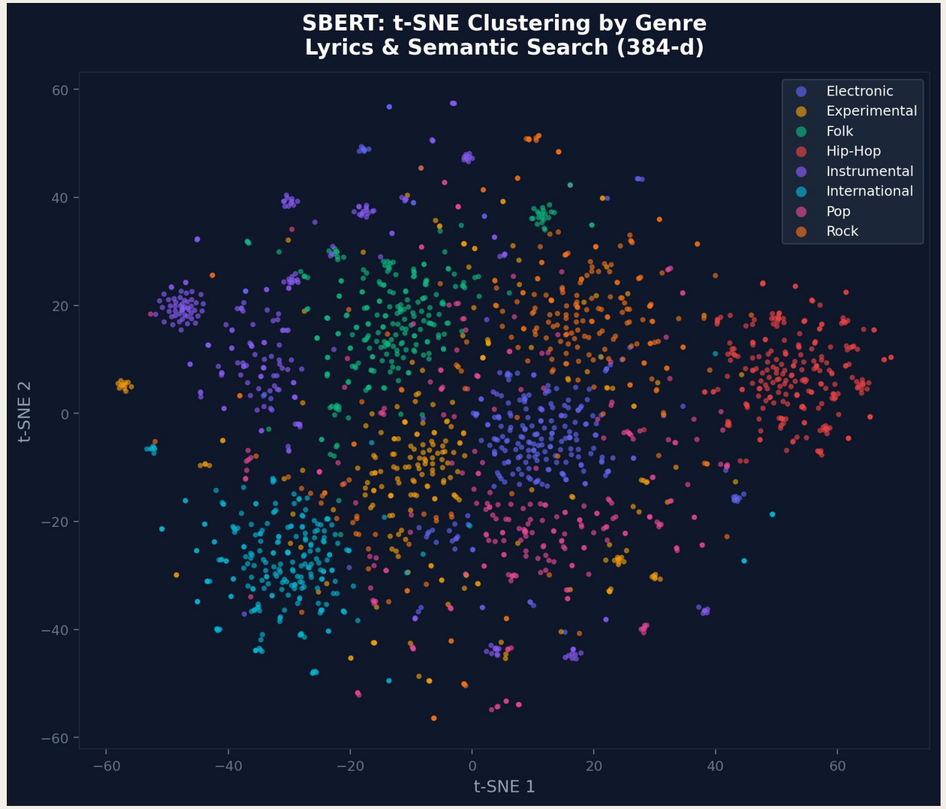
sentence-transformers

L3 **OpenL3** · self-supervised audio embeddings

marl/openl3

FAI **FAISS** · Facebook AI Similarity Search

facebookresearch/faiss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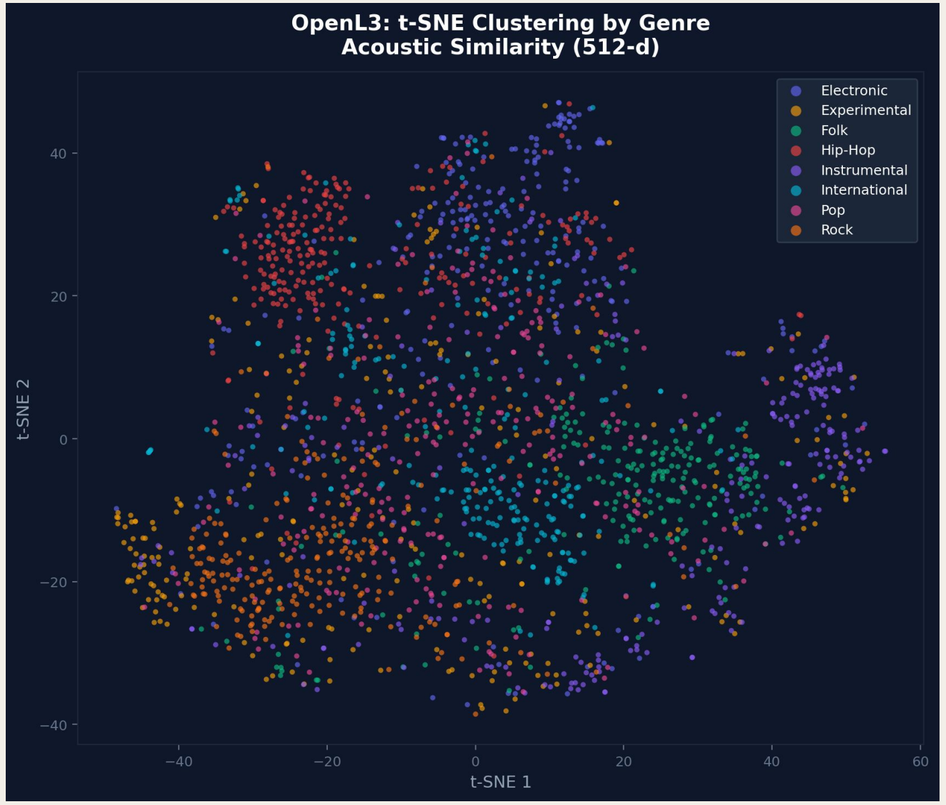
Lyrics & Semantic Search (384-d)
Electronic
Experimental
Folk
Hip-Hop
Instrumental
International
Pop
Rock
t-SNE 1
t-SNE 2
60
40
20
0
−20
−40
−60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Acoustic Similarity (512-d)
Electronic
Experimental
Folk
Hip-Hop
Instrumental
International
Pop
Rock
t-SNE 1
t-SNE 2
40
20
0
−20
−40
60

NOW PLAYING

Oh! Mighty Engine

Neil Halstead · FOLK

0:00 / 0:00

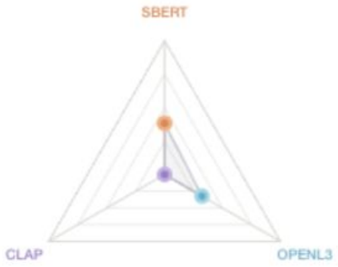
FUSED RECOMMENDATIONS

Reciprocal Rank Fusion

1	One Thing	Brett Saxon · FOLK	0.0400
2	A Penny For Your Dreams	Albin Andersson · FOLK	0.0287
3	In Your Hollow	Allysen Callery · FOLK	0.0282
4	Wayfaring Stranger	David Mumford · FOLK	0.0280
5	Train Coming In	Peter Biedermann · FOLK	0.0269
6	Tonight We Will Rise	Stephen Bartolomei · FOLK	0.0262
7	The Hermit's View	Itasca · FOLK	0.0245
8	Dust Devils	Michael Chapman · FOLK	0.0236
9	Walking In The Country	David Mumford · FOLK	0.0227
10	Merry-Go-Round	Howie & Ann Mitchell · FOLK	0.0164

Merry-Go-Round

Howie & Ann Mitchell



0.383	0.323	-0.095
SBERT	OPENL3	CLAP

SBERT · OPENL3 · CLAP

	SBERT	OPENL3	CLAP
1	Merry-Go-Round Howie & Ann Mitchell		0.383
2	Henry Martin Howie Mitchell		0.374
3	Irish Washerwoman Howie & Ann Mitchell		0.373
4	God speed the Plow The Fucked Up Beat		0.358
5	Living on the Wind Andy G. Cohen		0.332
6	Dona Nobis Pacem Howie Mitchell & Ruth Meyer		0.315

/ THANK YOU

QUESTIONS?

/ DEMO

/ SOURCE

github · multi-faceted-music-retrieval

/ TEAM

Wenny · Sid · Issac · MJ · Jiayi · Helena